

A
PROJECT REPORT
ON

PRAVA
College Academic Management System

Developed for
Women's College of Home Science and BCA, Loni

Submitted to
Savitribai Phule Pune University, Pune

Submitted by
Vaishnavi Vijay Kale | Roll No. 28

Under the Guidance of
Surekha Kale

For the Academic Year
2026–2027

In Partial Fulfilment of the Course
B.Sc (Computer Applications)
DEPARTMENT OF THE COMPUTER APPLICATIONS



Loknete Dr. Balasaheb Vikhe Patil (Padma Bhushan Awardee)

Pravara Rural Education Society's

Women's College of Home Science and BCA, Loni

Tal- Rahata, Dist- Ahilyanagar, Pin-413713

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Academic Year 2026–2027

PROJECT SUBMISSION

Submitted by

Vaishnavi Vijay Kale | Roll No. 28

Student of TY B.Sc (CA) – Third Year

B.Sc (Computer Applications)

I submit the group project

PRAVA

College Academic Management System

I, a student of TY B.Sc (CA), submit the group project entitled “PRAVA – College Academic Management System” for the academic year 2026–2027.

The project presents the design and implementation of an academic management portal for students, teaching staff and college administration. This report documents the application modules, database design, user interfaces and verification of the implemented system.

I have prepared this project as a group member under the guidance of Surekha Kale as part of the B.Sc (Computer Applications) project work at Savitribai Phule Pune University, Pune.

Place: _____

Date: _____

Vaishnavi Vijay Kale

Student signature | Roll No. 28

Loknete Dr. Balasaheb Vikhe Patil
(Padma Bhushan Awardee)

Pravara Rural Education Society's

Women's College of Home Science and BCA, Loni

CERTIFICATE

This is to certify that

Vaishnavi Vijay Kale | Roll No. 28

a student of TY B.Sc (CA) – Third Year, has prepared and submitted
the group project

PRAVA

College Academic Management System

under the guidance of Surekha Kale, in partial fulfilment
of the requirements for the degree of

B.Sc (Computer Applications)

Savitribai Phule Pune University, Pune

Academic Year 2026–2027

Dr. Sanjay Vani

Head of Department

External Examiner

Name and signature

Surekha Kale

Project Guide

Dr. Anushree Rajendra Khaire

Principal

Place: _____

College Seal

Date: _____

STUDENT DECLARATION

I, Vaishnavi Vijay Kale (Roll No. 28), a student of TY B.Sc (CA) at Women's College of Home Science and BCA, Loni, declare that this report presents my work on the group project “PRAVA – College Academic Management System” for the academic year 2026–2027.

The report describes the design and implementation of the college academic portal, including student, staff and office workspaces, database records, academic workflows and access controls. It is submitted under the guidance of Surekha Kale as part of my B.Sc (Computer Applications) project work.

College information, university curriculum documents, open-source software and technical references are acknowledged in the bibliography. Generated local records illustrate academic workflows; they do not certify college results. Submitted student profiles are retained separately from these presentation records.

Class: TY B.Sc (CA) – Third Year

Project Type: Group Project

Academic Year: 2026–2027

Place: _____

Date: _____

Vaishnavi Vijay Kale

Student signature | Roll No. 28

ACKNOWLEDGEMENT

I thank my project guide, Surekha Kale, for guidance in the preparation of my TY B.Sc (CA) project report, PRAVA – College Academic Management System.

I express my gratitude to Dr. Sanjay Vani, Head of Department, and Dr. Anushree Rajendra Khaire, Principal, for the academic environment in which this project is presented. I also thank the teaching staff of Women's College of Home Science and BCA, Loni for supporting my learning.

The project gave me an opportunity to connect web development with a practical college workflow. Preparing separate student, staff and office workspaces helped me understand how interface design, validation, database relationships and access permissions work together.

I acknowledge the college website, university syllabus documents, open-source software and official technical documentation used during the development and documentation of the project. These resources helped me describe the academic context and implement the application.

I thank my family, friends and everyone who assisted me with project review and demonstration preparation.

With sincere thanks,

Vaishnavi Vijay Kale | Roll No. 28

TY B.Sc (CA) – Academic Year 2026–2027

Technology Profile

PRAVA is a browser-based academic portal. A Python and Flask application manages accounts, validates academic operations and stores structured records in a relational database. The local presentation version uses SQLite and serves the website on the same computer.

Layer	Technology	Purpose
Presentation	HTML, CSS, JavaScript	Responsive public pages and role-specific workspaces
Templates	Jinja	Server-rendered pages with shared navigation
UI library	Bootstrap 5.3.3 and Icons 1.11.3	Locally bundled layout and interface assets
Application	Python 3.11–3.13; Flask 3.0.3	Routes, services and application configuration
Persistence	Flask-SQLAlchemy 3.1.1; SQLite	Models, relationships and local records
Authentication	Flask-Login 0.6.3; Werkzeug 3.0.3	Sessions and password-hash verification
Forms	Flask-WTF 1.2.1 and WTForms 3.1.2	Input validation and CSRF protection
Testing	Python unittest	Isolated automated functional and security checks
Delivery	GitHub and Windows launcher	Versioned source and repeatable local setup

The repository also contains PostgreSQL and optional external authentication configuration for future hosting. This report evaluates the working local-authentication installation. Package versions above describe the project requirements and bundled assets.

Abstract

PRAVA is a College Academic Management System for Women's College of Home Science and BCA, Loni. It connects public college information with student, staff and administrative workspaces. The project supports BCA and the Home Science specialisations in Food Science and Nutrition, and Textile Science and Designing, using a versioned 2024 NEP curriculum catalogue.

The website opens on Home. Its header contains Home, Academics, Contact and a Login dropdown. Student login continues through BCA or Home Science selection, while Staff and Office / Administration open their own forms. The authenticated account determines the displayed name, permissions and dashboard. Night mode is the default, with a Day mode option.

Students can submit a website sign-up request for Office approval. Separate Google Forms collect student and staff profile information into private response spreadsheets. Those responses require verification and conversion to the portal CSV templates; submitting a Google Form does not automatically create a PRAVA account.

Python, Flask, Jinja and SQLAlchemy implement the application, while a local SQLite database preserves users and academic records. The 22 model tables support profiles, teaching allocations, attendance, marks, materials, assignments, notices, curriculum sources, activities and review workflows. Password hashing, CSRF protection, role checks and record ownership restrict access.

The repository baseline is 4abb237 with local typography improvements. All 63 automated tests passed on 5 October 2026. This report documents the implementation, database, academic workspaces and setup. Final-year syllabus verification, official NEP grade calculations and account notifications remain future enhancements.

Keywords: academic management, Flask, SQLite, BCA, Home Science, NEP curriculum, role-based access.

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1 Introduction Objectives and Scope

College academic records connect identities, programme enrolment, teaching allocations, attendance, assessments and learning resources. Students need access to their own progress; teachers need the records of assigned classes; and the Office needs consistent institutional data. PRAVA brings these responsibilities into one browser-based application.

Problem and proposed system

Keeping the same academic information in separate registers, files and messages can make retrieval slow and introduce duplicate entries. PRAVA addresses this design problem through shared relational records and server-side access rules. Public college content, pending registrations and verified academic profiles are maintained separately.

Objectives

- Provide clear public navigation and role-specific login with the correct account identity.
- Verify student registration and roster imports before activating academic accounts.
- Link student and staff work to the correct programme, curriculum, semester and subject.
- Manage attendance, marks, materials, assignments, notices, activities and reports.
- Provide source-linked syllabus information and a repeatable local demonstration setup.

Project scope

The catalogue covers BCA, B.Sc. Food Science and Nutrition, and B.Sc. Textile Science and Designing. FY, SY and TY navigation represents the three-year journey; detail is displayed according to the available source review. Office and Principal access use distinct administrative permissions. The local project demonstrates academic operations; fees, payroll, library circulation and university result certification are outside its current modules.

2 Requirements Analysis and Feasibility

The requirements come from the requested college workflows, the supplied report format, college and curriculum references, and source-code inspection. Google Forms support profile collection. Submitted records require Office review; the project does not claim measured adoption benefits or institutional-scale operation.

Requirement	Local demonstration setup
Hardware	64-bit laptop or desktop; 4 GB RAM and 1 GB free space are planning recommendations, not measured minima.
Software	Python 3.11–3.13, a modern browser and the pinned requirements.txt dependencies.
Database and files	SQLite in instance/prava.sqlite3; learning files and profile uploads stored separately.
Connectivity	Internet for first setup and external sources. Bundled local pages render after setup without internet.
Quality	Readable Day/Night themes, validation feedback, persistent records, controlled access and auditable review.

Feasibility

Technical feasibility follows from the conventional browser, Flask and relational database architecture. SQLite supports local presentation without a separate database server. Operational feasibility depends on the Office verifying identities and allocating subjects before Staff enters records. Separate dashboards make these responsibilities visible.

An existing laptop and open-source dependencies support an economical local demonstration. Printing, binding and any later hosting remain separate costs. Data feasibility depends on official student identifiers, staff assignments and applicable batch documents. Missing academic results remain pending, while unresolved curriculum items retain review status.

3 System Architecture and Modules

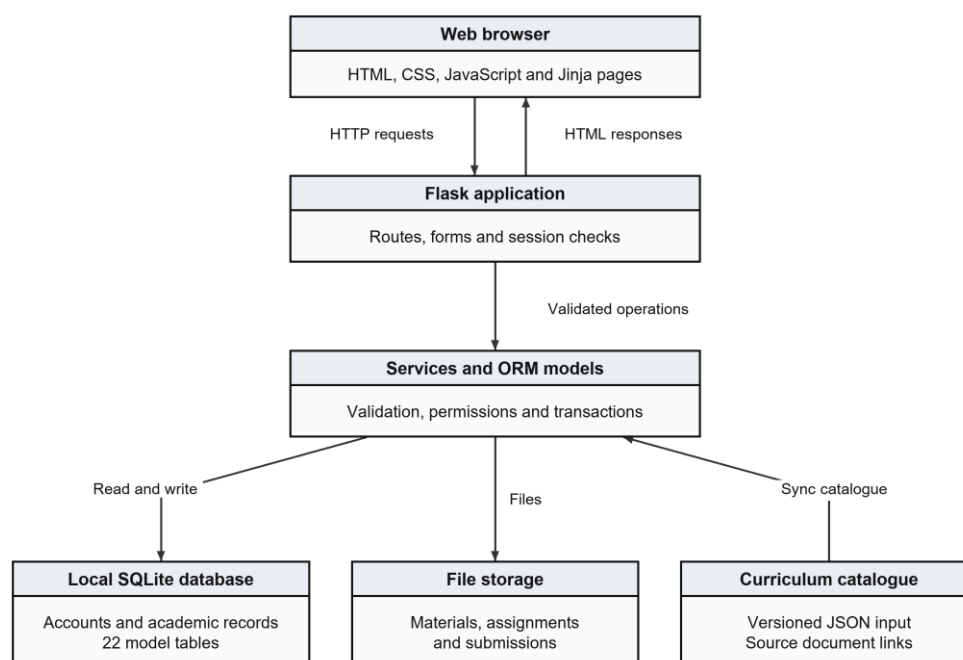


Figure 1 Browser application services and local storage

Flask routes receive requests, validate forms and check sessions and roles. Services perform academic operations through SQLAlchemy. Jinja renders HTML responses; uploaded files remain separate from their metadata. The curriculum JSON synchronises reference information into the database.

Module	Principal responsibility
Public and access	Home, college information, academics, contact, role login and student sign-up.
Student and Staff	Programme-based progress views; assigned-subject attendance, marks and learning resources.
Administration	Profiles, approvals, curriculum review, imports, notices, access scopes and exports.
Shared services	Database transactions, validation, activity records, upload checks and persistent setup.

4 Login Registration and Permissions

Entry point	Destination and rule
Home → Login dropdown	Student, Staff and Office / Administration appear immediately below Login.
Student → BCA / Home Science	Dedicated form validates credentials, active status, role and programme.
Staff	Staff credentials open the faculty workspace and assigned-subject tools.
Office / Administration	Administrative workspace follows Office, Principal or administrator scope.
Back Switch and Logout	Back uses defined internal pages. Switching requires new credentials. Logout uses protected POST.

Student sign-up and approval

Website sign-up captures identity, programme, semester, admission year and a password that is stored only as a hash. A valid request becomes Pending in student_registrations. Office approval rechecks duplicate identities and creates the user and student profile in one transaction. Rejection preserves the review decision without activating an account.

Permission boundaries

A student reads her own academic records. Staff works with explicitly assigned subjects and eligible students. Office manages authorised records and approvals. Principal scope is read-only for permitted oversight. Privileged account creation requires administrator scope. A failed account switch preserves the earlier authenticated identity; choosing a portal alone does not change the account.

Public student sign-up currently uses local password authentication. Optional external authentication requires additional provisioning work. Registration approval emails are not automated.

Use Case Diagram

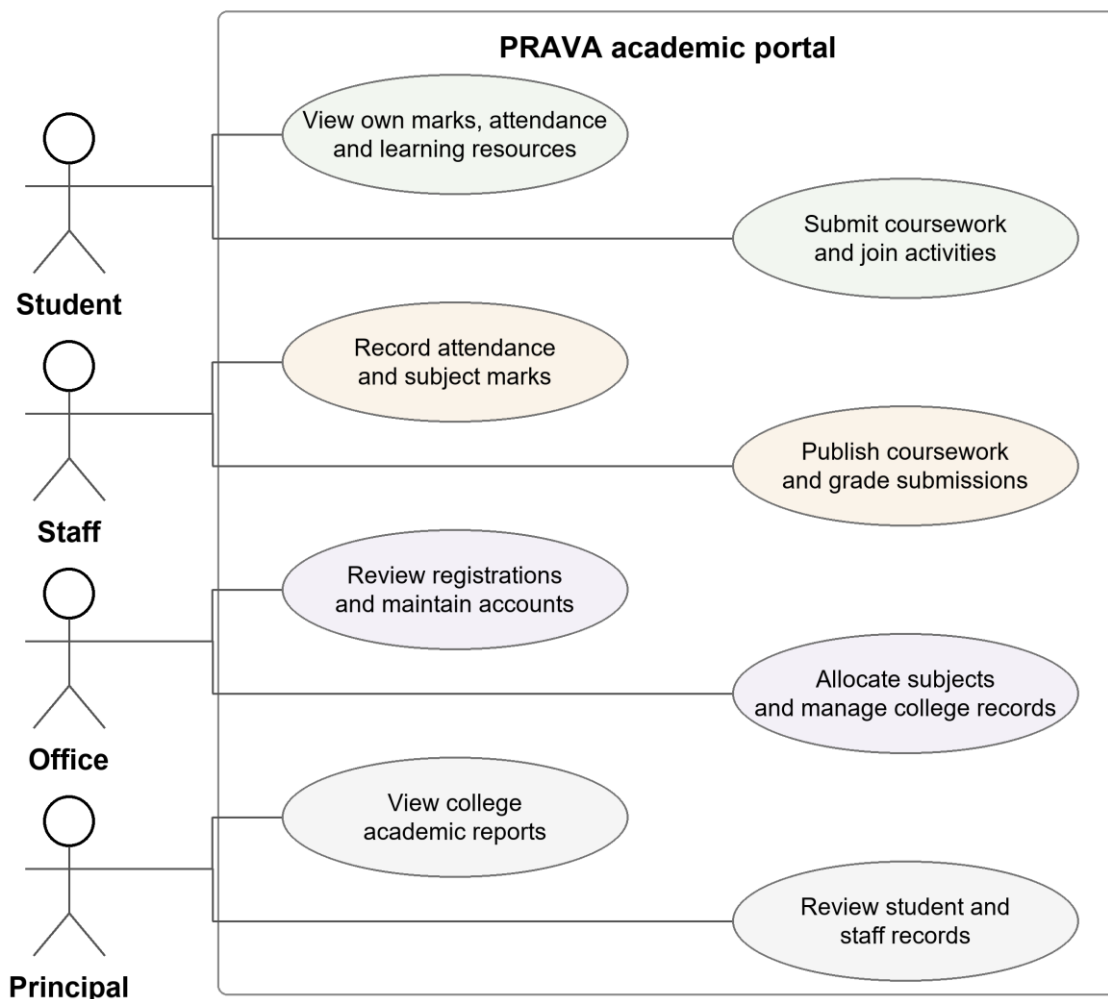


Figure 2 Primary actors and their academic operations

Student access is restricted to her own records. Staff operations depend on subject allocation. Office handles record maintenance and registration review; the Principal account provides read-only oversight.

Protected operations require an active authenticated account. A prospective student may submit a public sign-up request; Office approval is required before that account can access the student workspace.

5 Database Design and ER Diagrams

The implementation contains 22 model tables and 227 columns. PK denotes a primary key, FK a foreign key and UQ a uniqueness constraint. A 1-to-many relationship is implemented by a child foreign key; 1-to-0..1 describes an optional unique profile for a user.

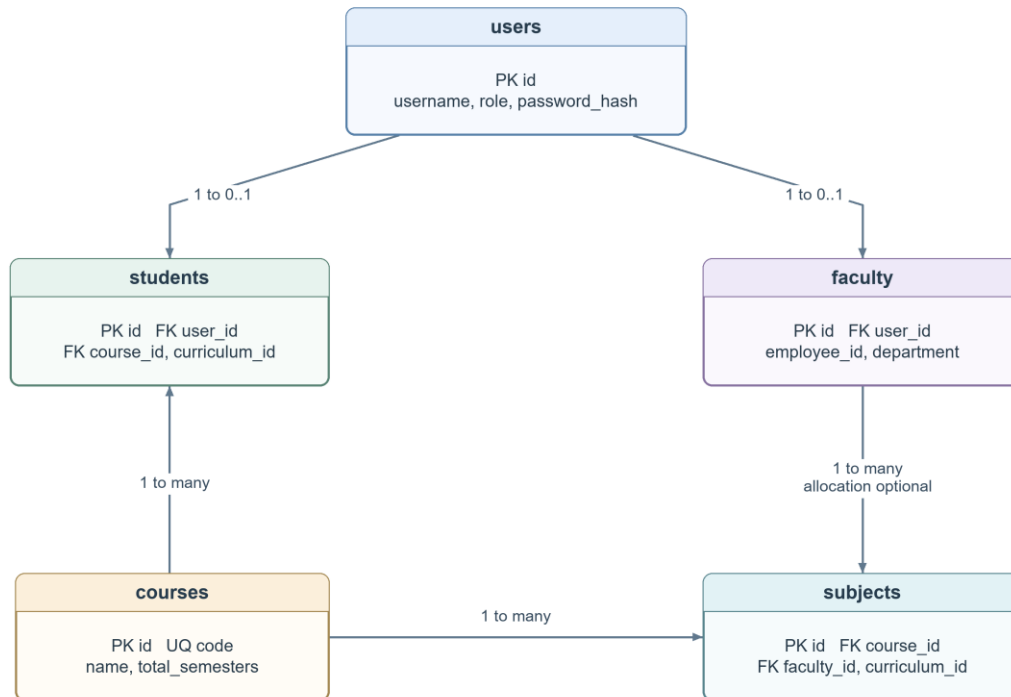


Figure 3 Accounts profiles courses and teaching subjects

users holds authentication identity; students and faculty hold role-specific profiles. Students belong to courses and curricula. subjects describes actual teaching allocations, including programme, semester and faculty. The separate curriculum catalogue provides source-reviewed reference entries.

The data dictionary follows SQLAlchemy declarations. Application validation and SQLite enforcement can differ. Password fields contain hashes; JSON text fields hold structured data. Foreign-key relationships are shown only where declared.

Academic and Learning Relationships

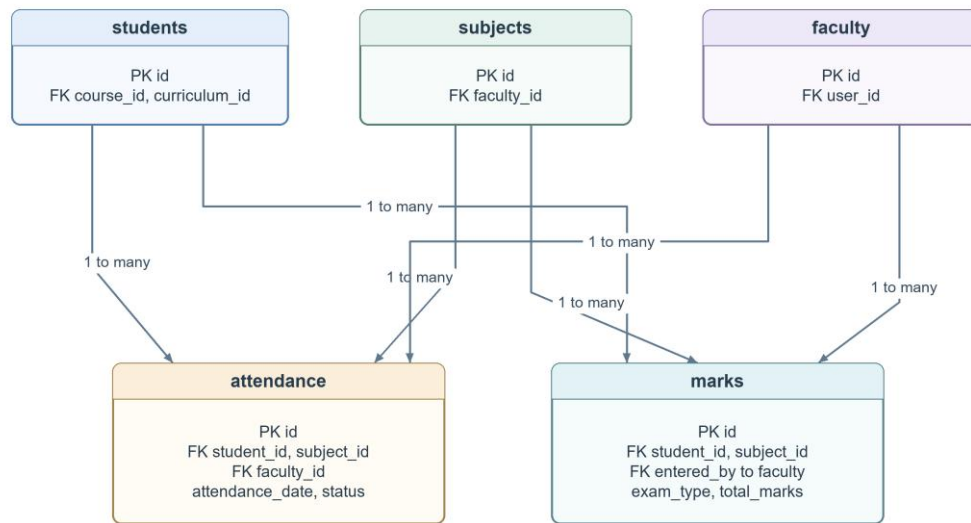


Figure 4 Student subject and faculty references in assessment

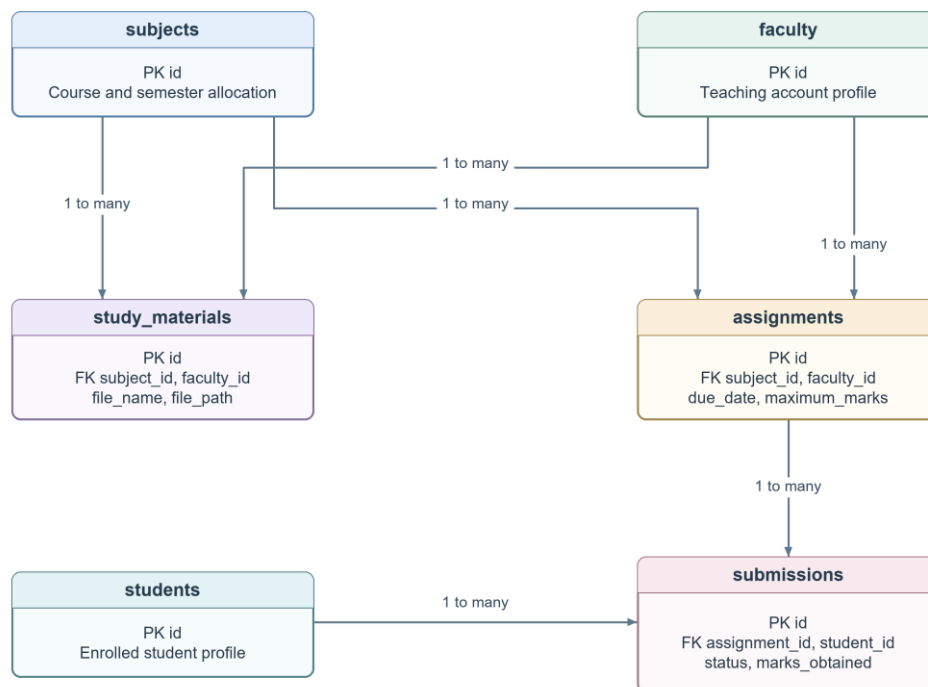


Figure 5 Learning materials assignments and submissions

Attendance is unique per student, subject and date; marks per student, subject and exam type. An assignment has at most one tracked submission per student. All academic operations also check the authorised subject relationship.

Curriculum and Participation Relationships

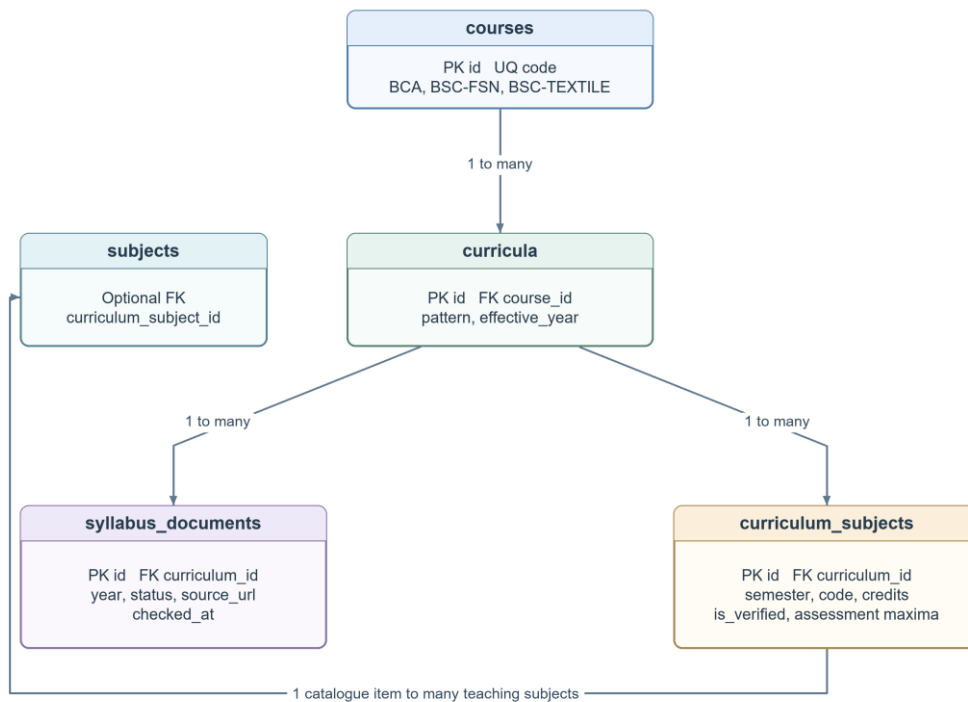


Figure 6 Versioned curriculum documents and catalogue subjects

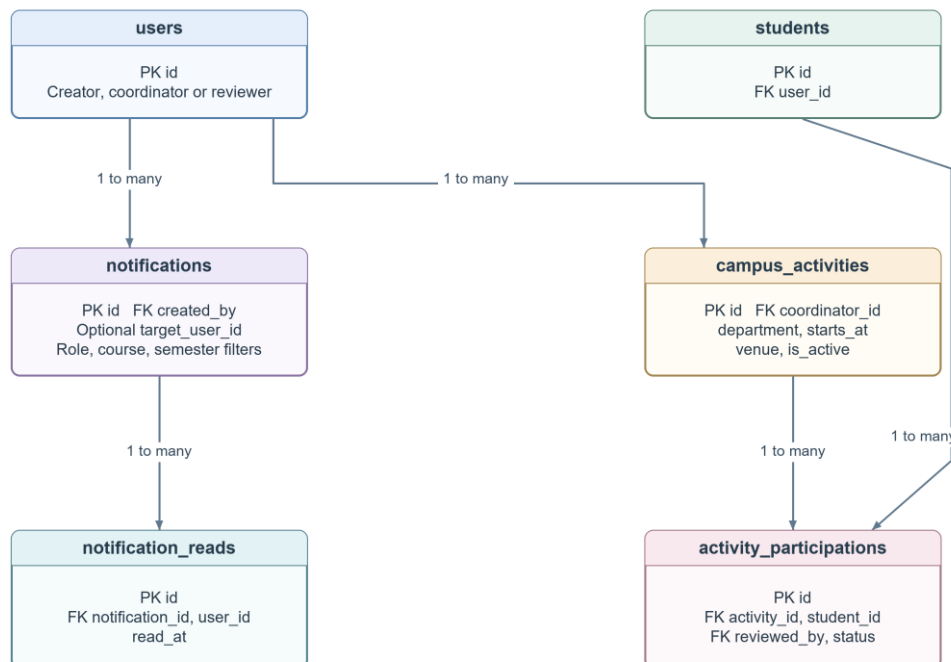


Figure 7 Notifications read receipts and activity participation

Source documents and catalogue subjects belong to a curriculum pattern. Activity participation links a student to an activity and its review. Contact inquiries and student-test snapshots are independent records; their models do not declare a student-account foreign key.

6 Data Flow Diagrams

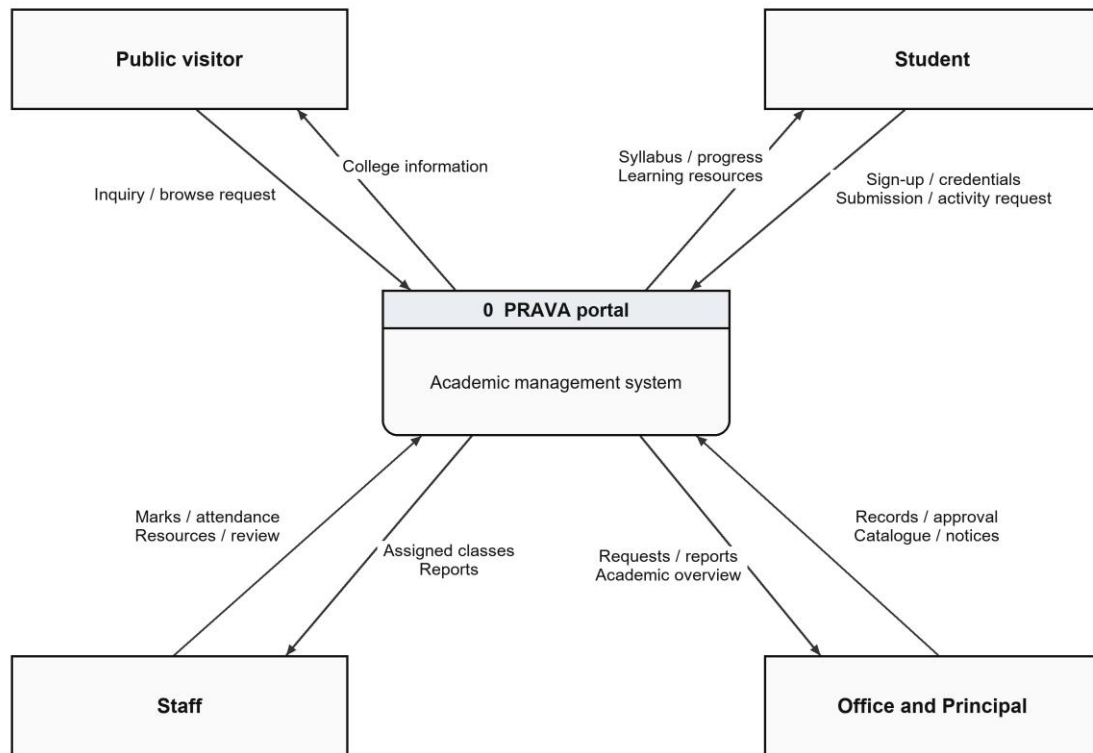


Figure 8 Context DFD for the PRAVA academic portal

The context diagram shows the system boundary. Visitors request public information and submit inquiries. Students supply registration and learning inputs and receive their academic views. Staff maintain assigned academic records. Office manages approvals and institutional records; Principal access provides permitted oversight.

Arrows identify information exchanged with the system. The database is internal to PRAVA and is expanded into logical stores at the next level. Google Forms collection is a separate preparatory intake process, followed by verified CSV import.

Level One Data Flow

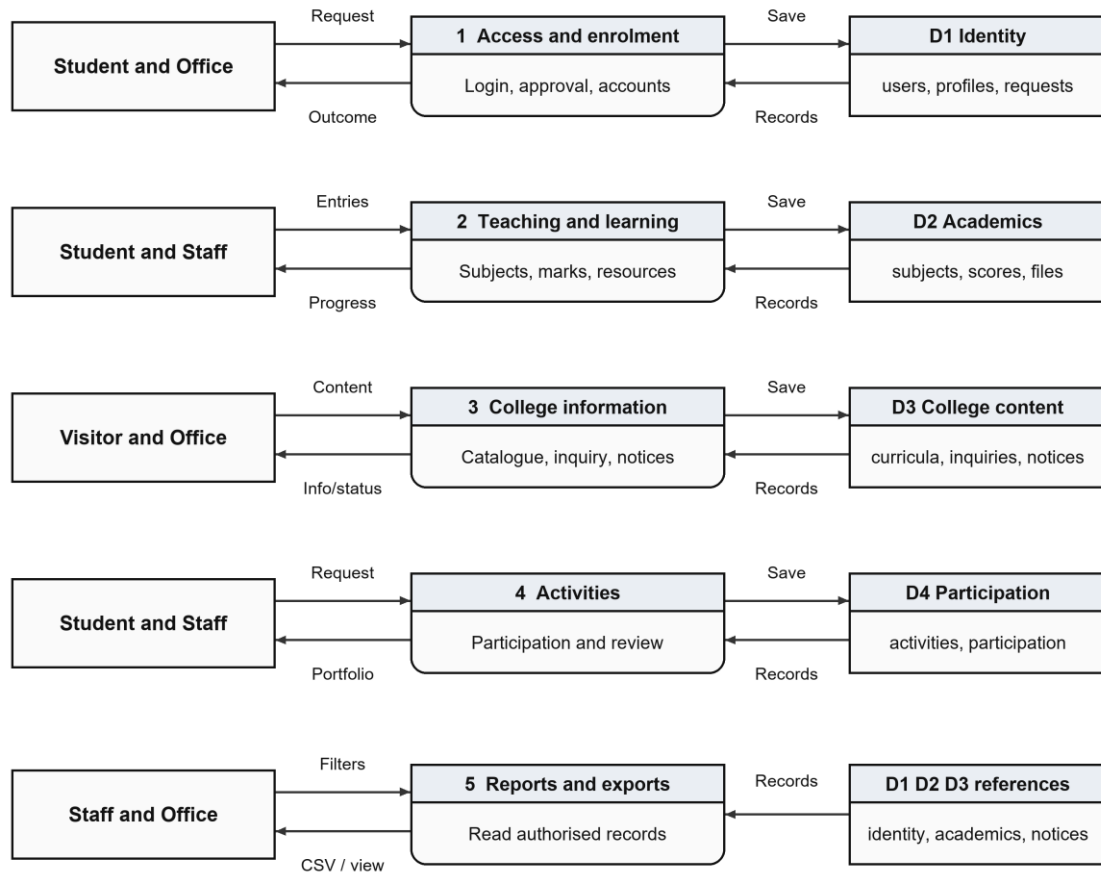


Figure 9 Level one decomposition and logical data stores

The five processes cover access and enrolment, teaching and learning, college information, activities, and reporting. D1–D4 group related tables in the same relational database. Reporting reads existing records within the current account scope and produces views or CSV files.

A report does not create a second independent academic total. The data stores are logical groups; the application maintains a shared schema with database keys and service-level validation.

Level Two Registration Review

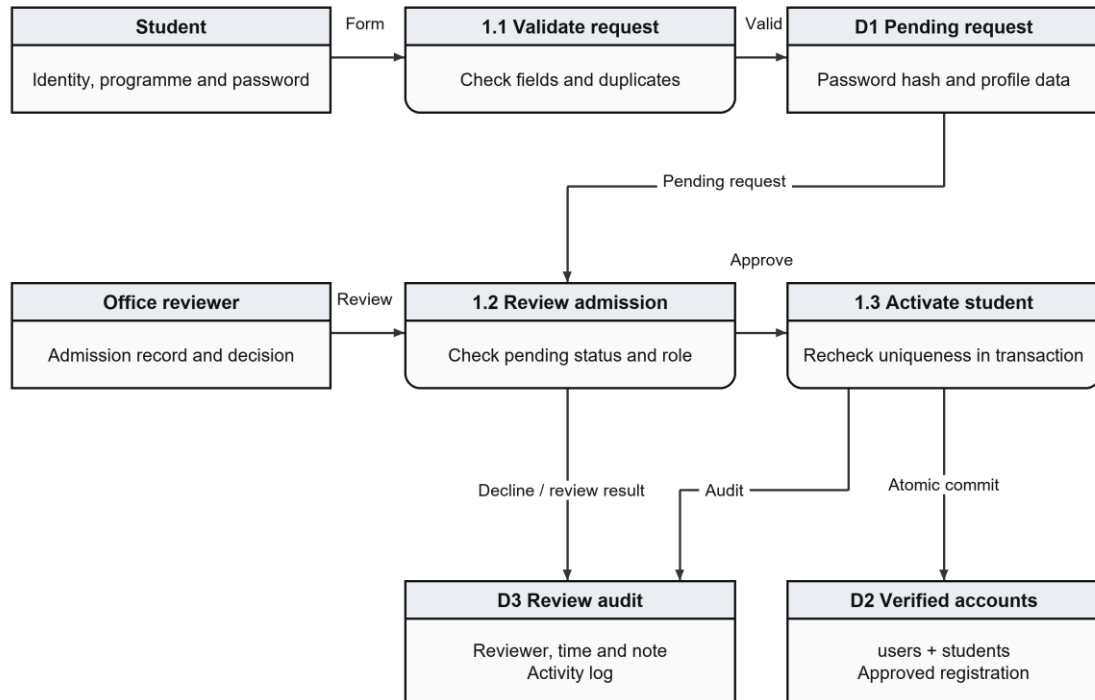


Figure 10 Student sign-up validation Office review and activation

A valid website sign-up creates a pending request after format and duplicate checks. An authorised Office user reviews the request. Approval repeats the uniqueness checks and creates a login identity and student profile atomically. A conflict rolls the operation back and returns review feedback.

The review records its decision, reviewer and time. Rejected requests remain reviewed. Approval permits the student to sign in with the chosen credentials; it does not automatically send an email or create subject allocations.

7 Data Dictionary

The schema contains 22 tables and 227 declared columns. This five-page summary lists every table and selected fields needed to explain the principal workflows. Repeated audit timestamps and secondary profile fields remain defined in the source models.

Table	Purpose
users	Login identity, role, scope and active status.
students	Student enrolment and programme profile.
faculty	Staff identity and professional profile.
courses	Degree programmes and semester count.
subjects	Teaching subjects and faculty allocation.
attendance	Date-wise student attendance.
marks	Subject assessments and component scores.
assignments	Faculty coursework and due dates.
submissions	Student work, marks and feedback.
study_materials	Uploaded learning-resource metadata.
student_registrations	Pending sign-ups and Office decisions.
curricula	Programme pattern and effective year.
curriculum_subjects	NEP catalogue subjects and assessment limits.
syllabus_documents	Year-wise source links and review status.
campus_activities	College activities and coordinators.
activity_participations	Student activity registration and review.
notifications	Targeted announcements and expiry.
notification_reads	Per-user notice read receipts.
contact_inquiries	Public contact messages and status.
record_imports	Staged CSV import payload and owner.
activity_logs	Recorded user actions and timestamps.
student_test_responses	Student test answers, score and feedback.

Notation: PK = primary key; FK = foreign key; UQ = unique. Each table has an id primary key. Field types and keys below follow the SQLAlchemy declarations; validation rules are described separately.

Identity and Enrolment

Table	Field	Type	Key / meaning
users	id	INTEGER	PK
	username	VARCHAR(80)	UQ
	full_name	VARCHAR(120)	Account holder name
	email	VARCHAR(120)	UQ
	password_hash	VARCHAR(255)	One-way password hash
	role	VARCHAR(20)	student / faculty / admin
	admin_scope	VARCHAR(30)	Administrator / office / principal scope
	is_active	BOOLEAN	Controls account availability
students	id	INTEGER	PK
	user_id	INTEGER	UQ; FK users.id
	enrollment_number	VARCHAR(50)	UQ
	course_id	INTEGER	FK courses.id
	curriculum_id	INTEGER	FK curricula.id
	semester	INTEGER	Current / assigned semester
	admission_year	INTEGER	Year of admission
faculty	id	INTEGER	PK
	user_id	INTEGER	UQ; FK users.id
	employee_id	VARCHAR(50)	UQ
	department	VARCHAR(120)	Teaching department
	qualification	VARCHAR(120)	Recorded qualification

users stores shared authentication identity. A unique user_id connects each student or faculty profile to one account. Enrolment numbers and employee IDs identify the corresponding college records.

Teaching and Assessment

Table	Field	Type	Key / meaning
courses	id	INTEGER	PK
	code	VARCHAR(30)	UQ
	name	VARCHAR(120)	Display name
	total_semesters	INTEGER	Programme semester count
subjects	id	INTEGER	PK
	code	VARCHAR(30)	Programme / subject code
	course_id	INTEGER	FK courses.id
	semester	INTEGER	Current / assigned semester
	faculty_id	INTEGER	FK faculty.id
	curriculum_subject_id	INTEGER	FK curriculum_subjects.id
attendance	id	INTEGER	PK
	student_id	INTEGER	FK students.id
	subject_id	INTEGER	FK subjects.id
	attendance_date	DATE	Date of the class record
	status	VARCHAR(20)	Workflow / attendance state
marks	id	INTEGER	PK
	student_id	INTEGER	FK students.id
	subject_id	INTEGER	FK subjects.id
	exam_type	VARCHAR(50)	Assessment label
	internal_marks	INTEGER	Internal component
	external_marks	INTEGER	External component
	total_marks	INTEGER	Internal + external total

Composite uniqueness: subjects (code, course_id, semester); attendance (student_id, subject_id, attendance_date); marks (student_id, subject_id, exam_type). Assessment services check eligibility and configured component limits.

Coursework and Registration

Table	Field	Type	Key / meaning
assignments	id	INTEGER	PK
	subject_id	INTEGER	FK subjects.id
	faculty_id	INTEGER	FK faculty.id
	due_date	DATETIME	Submission deadline
	maximum_marks	INTEGER	Maximum assignment score
submissions	id	INTEGER	PK
	assignment_id	INTEGER	FK assignments.id
	student_id	INTEGER	FK students.id
	marks_obtained	INTEGER	Awarded assignment score
	faculty_feedback	TEXT	Review comments
	status	VARCHAR(30)	Workflow / attendance state
study_materials	id	INTEGER	PK
	subject_id	INTEGER	FK subjects.id
	file_path	VARCHAR(255)	Stored resource path
student_registrations	id	INTEGER	PK
	enrollment_number	VARCHAR(50)	UQ
	curriculum_id	INTEGER	FK curricula.id
	status	VARCHAR(20)	Workflow / attendance state
	reviewed_by	INTEGER	FK users.id
	user_id	INTEGER	UQ; FK users.id

One submission is tracked per assignment and student. Registration review activates an approved account and links user_id; approval rechecks uniqueness. Uploaded-file paths identify controlled local storage, not a second academic database.

Curriculum and Activities

Table	Field	Type	Key / meaning
curricula	id	INTEGER	PK
	course_id	INTEGER	FK courses.id
	pattern	VARCHAR(80)	Curriculum pattern label
syllabus_documents	id	INTEGER	PK
	curriculum_id	INTEGER	FK curricula.id
	year	INTEGER	Programme year
	source_url	VARCHAR(1000)	Reference document URL
	status	VARCHAR(30)	Workflow / attendance state
curriculum_subjects	id	INTEGER	PK
	curriculum_id	INTEGER	FK curricula.id
	semester	INTEGER	Current / assigned semester
	code	VARCHAR(60)	Programme / subject code
	credits	NUMERIC(4, 1)	Published credit value
	is_verified	BOOLEAN	Catalogue verification flag
campus_activities	id	INTEGER	PK
	title	VARCHAR(180)	Activity title
	coordinator_id	INTEGER	FK users.id
activity_participations	id	INTEGER	PK
	activity_id	INTEGER	FK campus_activities.id
	student_id	INTEGER	FK students.id
	status	VARCHAR(30)	Workflow / attendance state
	reviewed_by	INTEGER	FK users.id

Composite uniqueness: curriculum (course_id, pattern); syllabus document (curriculum_id, year); catalogue subject (curriculum_id, semester, code); participation (activity_id, student_id). Catalogue verification and activity review remain explicit recorded states.

8 Academic Processing and Reports

Subject allocation and attendance

Office creates teaching subjects for a course, curriculum and semester, then assigns the responsible faculty member. Staff receives the eligible cohort for that allocation. Attendance stores Present, Absent or Late per student and class date. The current report counts Present and Late as attended sessions.

Attendance percentage = $\text{attended records} \div \text{recorded sessions} \times 100$. With no sessions, progress remains pending. The implemented 75 percent warning is an application threshold that requires college confirmation before formal use.

Marks and assignments

Internal and external marks must be non-negative and within configured maxima. NEP entry also requires a verified assessment link and matching component limits. The total is internal plus external marks; official NEP grade, SGPA and CGPA conversion is pending. Assignments store deadlines and maximum marks; submissions hold work, status and faculty feedback.

Report	Information produced
Students and Faculty	Identity, programme or department, contact and active status.
Attendance	Student, subject, date, status and recording faculty.
Marks	Configured scores, totals and existing generic threshold-based result.
Assignments	Assignment, subject, due date, maximum marks and submission count.
Notifications	Title, audience, creator, active status and timing.

CSV and print-friendly views support review of stored records. The generic pass/fail export uses the subject passing threshold; it is not an official university result calculation.

9 Student and Staff Information Collection

Separate English Google Forms collect Student and Staff profiles into private response spreadsheets. As of 5 October 2026, nine student responses represent eight distinct BCA profiles after duplicate review. The local portal contains 19 students: eight submitted profiles and 11 generated identities, including five newly added Home Science students and FY/SY BCA students. Eleven staff profiles have teaching allocations.

Form	Required information	Optional information
Student	Full name, email, college Enrollment/PRN, programme, class, semester, admission year and confirmation.	Mobile, date of birth, gender and address.
Staff	Full name, email, Employee ID, department, teaching subjects with programme/semester, and confirmation.	Qualification, mobile and joining date.

Verified entry into PRAVA

- Collect responses using the relevant link. Passwords are not collected in Google Forms.
- Office checks identifiers, email uniqueness, programme, semester and admission year against official records.
- Prepare the exact Student or Staff UTF-8 CSV template, assigning a unique username to each verified person.
- Validate the import preview, then save the records. Imported accounts remain inactive until individual passwords and activation are completed by Office.
- Enter optional profile fields and teaching allocations separately. Student website sign-up remains a separate approval workflow.

Programme codes: BCA, BSC-FSN and BSC-TEXTILE; pattern: 2024 NEP. The local dataset includes 135 teaching subjects. Generated attendance, marks and coursework are linked only to presentation identities. Submitted profile values are preserved as authorised; unresolved details remain subject to college verification.

10 Input and Output Screens

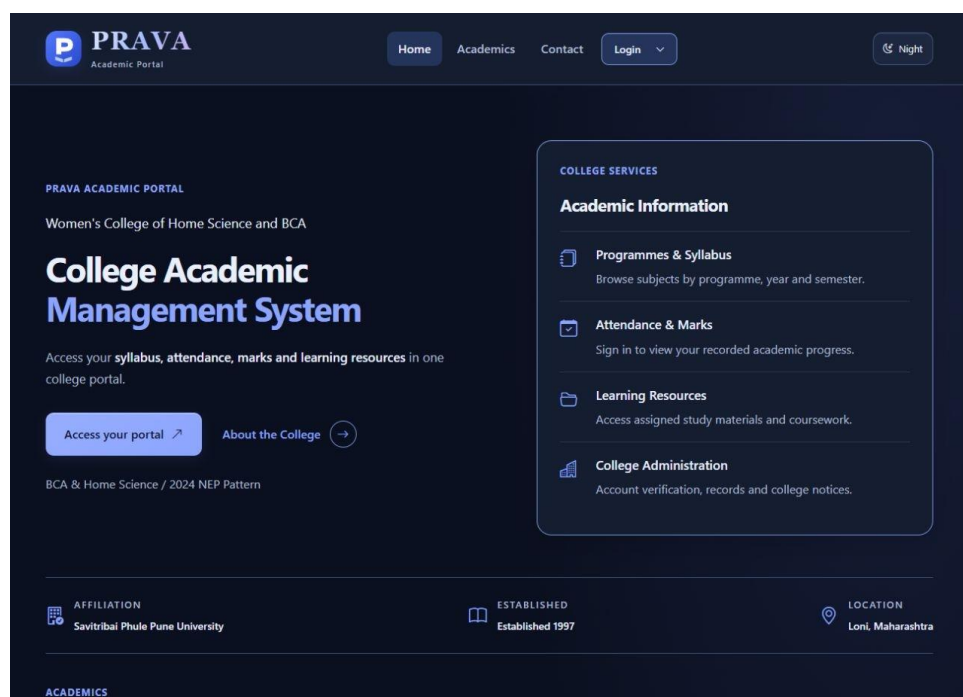


Figure 11 Current Home page in default Night mode

The header presents Home, Academics, Contact and Login. The revised heading identifies the College Academic Management System. Day mode is available from the theme control. The homepage presents academic services, programme information and college notices with functional headings.

Screenshots were refreshed on 5 October 2026 after the readability update. Named accounts display academic workflows. Generated academic entries illustrate local operation and are not official college results; the eight submitted BCA profiles retain their separate intake provenance.

Login Dropdown and Programme Selection

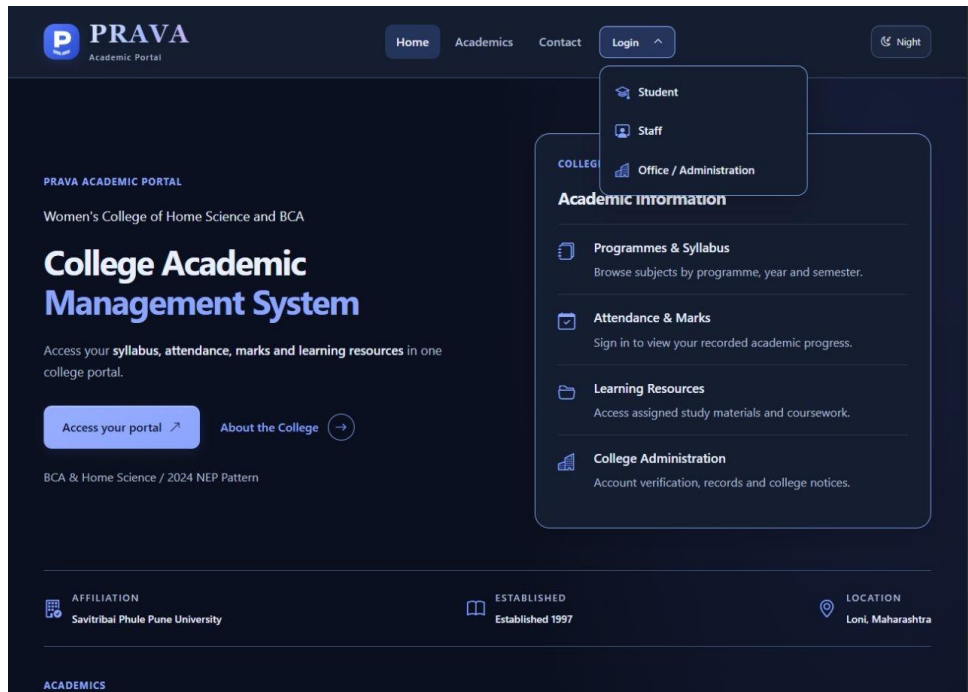


Figure 12 Login dropdown with Student Staff and Office options

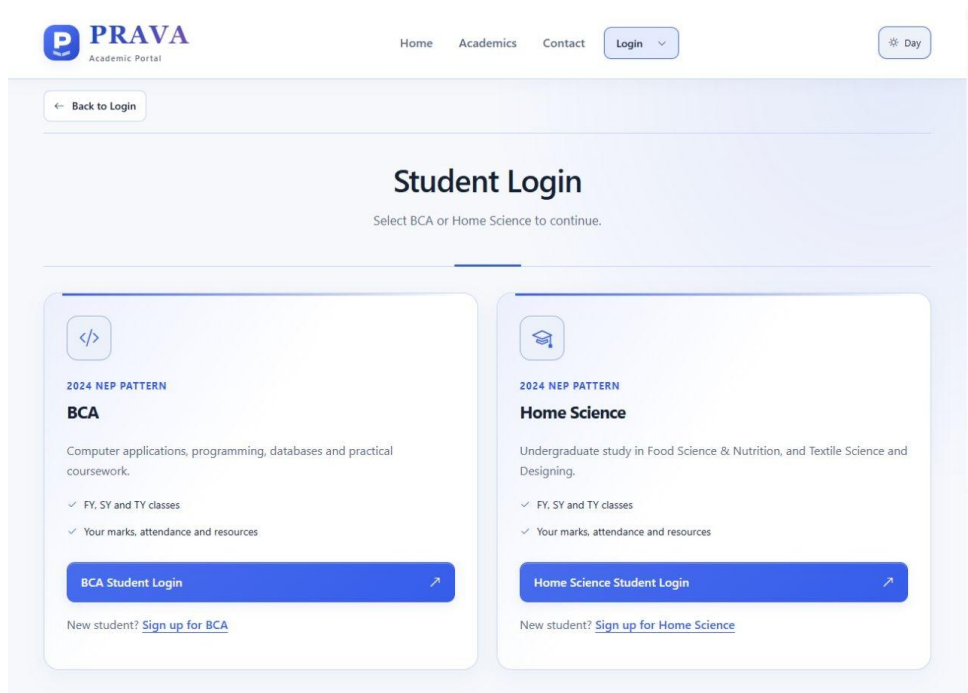


Figure 13 BCA and Home Science programme selection

Login opens a dropdown directly below the header control. Student continues to programme selection; the programme pages retain student sign-up links.

Student Login and Website Sign Up

The screenshot shows the PRAVA Academic Portal interface. At the top, there is a navigation bar with the PRAVA logo, links for Home, Academics, Contact, and a Login button. A 'Day' button is also present. Below the navigation bar, there is a 'Back to programmes' link. The main content area is divided into two sections. The left section, titled 'PRAVA ACADEMIC PORTAL', features a dark blue background with a graduation cap icon and the text 'BCA Student Login'. It also includes a sub-header 'View your attendance, marks, subjects and assignments.' The right section, titled 'BCA STUDENT PORTAL', contains a 'Sign In' form. The form includes fields for 'Username or Email' and 'Password', a 'Remember Me' checkbox, and a 'Need help?' link. A blue 'Sign in as BCA Student' button is at the bottom of the form. Below the button, there is a link for 'New to PRAVA? Create your student account' and a note stating 'Access is linked to your college account, programme and role.'

Figure 14 BCA student login form

The screenshot shows the PRAVA Academic Portal Student Registration form. The form is titled 'Student Registration' and includes a sub-header 'Join as a BCA or Home Science student. The college office checks your enrollment before activating your account.' The form is divided into two main sections. The left section, titled 'Registration Process', contains three steps: 1. 'Your details' (Use your own email and college enrollment / PRN), 2. 'Office verification' (The office checks your programme, semester and enrollment), and 3. 'Account access' (Once approved, sign in with your chosen username and password). It also includes a note 'Staff accounts are issued by the Administration office.' The right section contains the registration form fields: 'Full name', 'Choose a username', 'Your email', 'College enrollment / PRN', 'Your programme' (dropdown menu), 'Current semester' (dropdown menu), 'Admission year', 'Create password', and 'Confirm password'. A blue 'Submit student sign-up' button is at the bottom of the form. Below the button, there is a note: '2024 NEP / Your password must contain at least 8 characters. Sign-up details are stored securely for Office review.'

Figure 15 Student sign-up form awaiting Office verification

The role and programme are explicit on login. Website sign-up stores a pending request; successful submission alone does not grant active account access.

BCA Student Dashboard and Syllabus

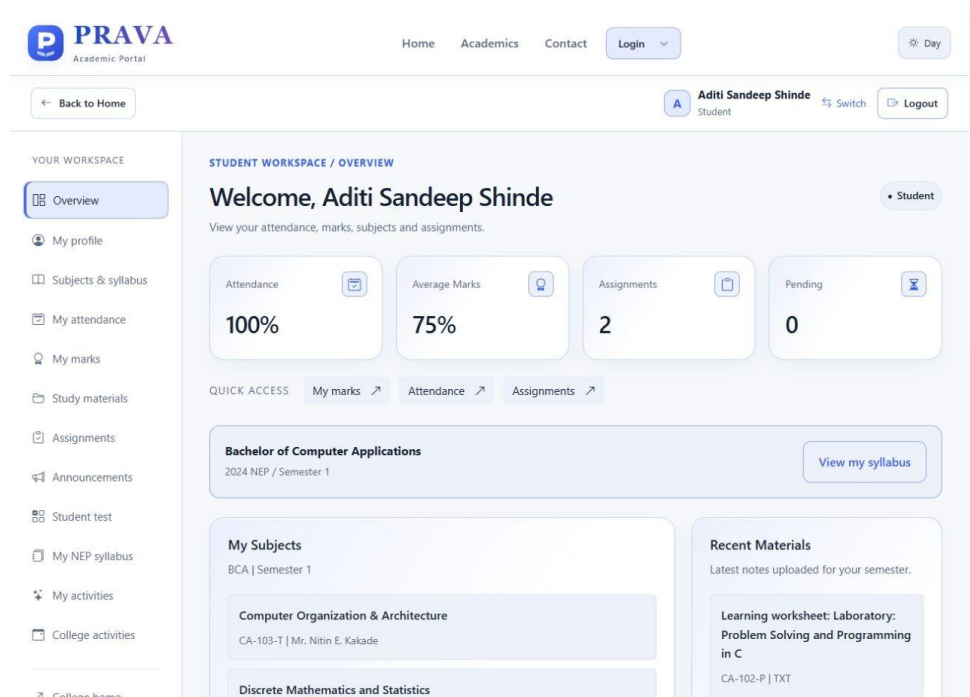


Figure 16 BCA student dashboard

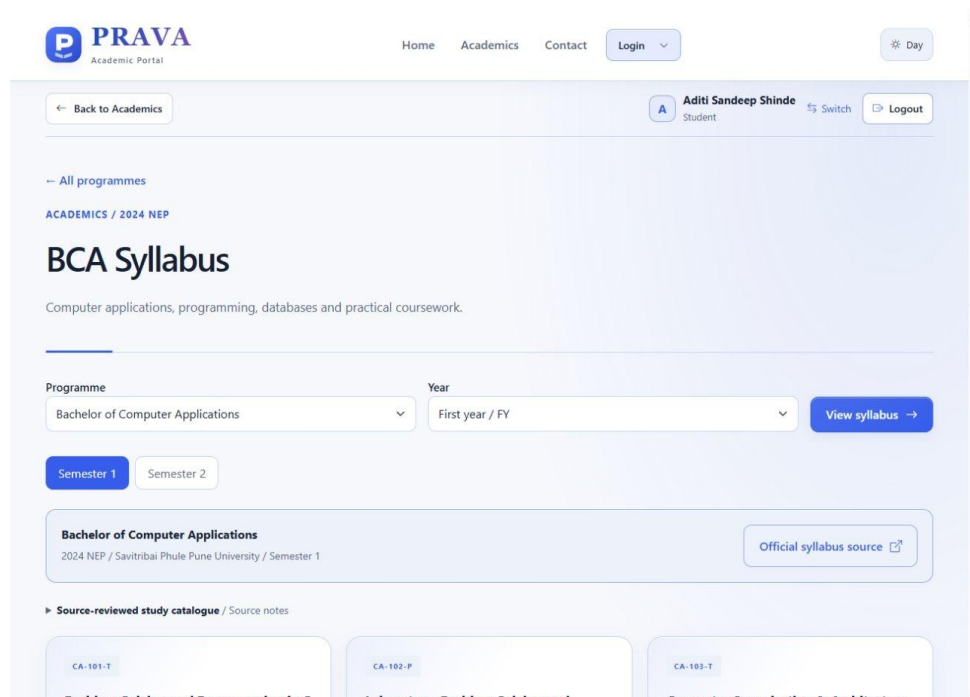


Figure 17 BCA NEP catalogue and official source link

Aditi Sandeep Shinde is the configured BCA presentation account. Its dashboard shows attendance, marks, coursework and allocated teaching subjects. The syllabus screen retains the source-linked NEP reference catalogue.

Home Science Dashboard and Syllabus

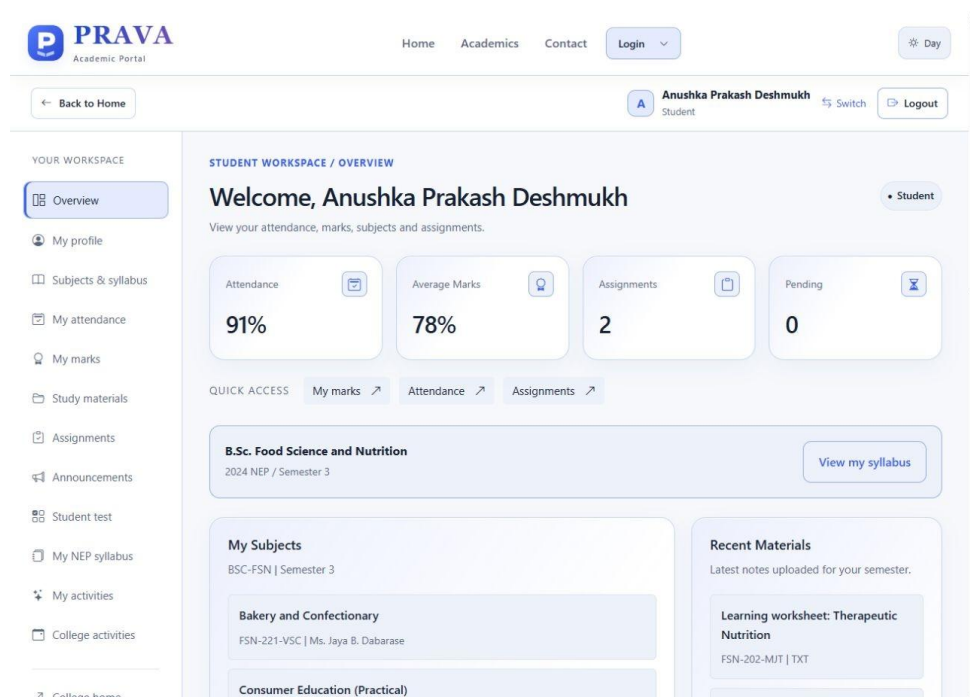


Figure 18 Home Science student dashboard

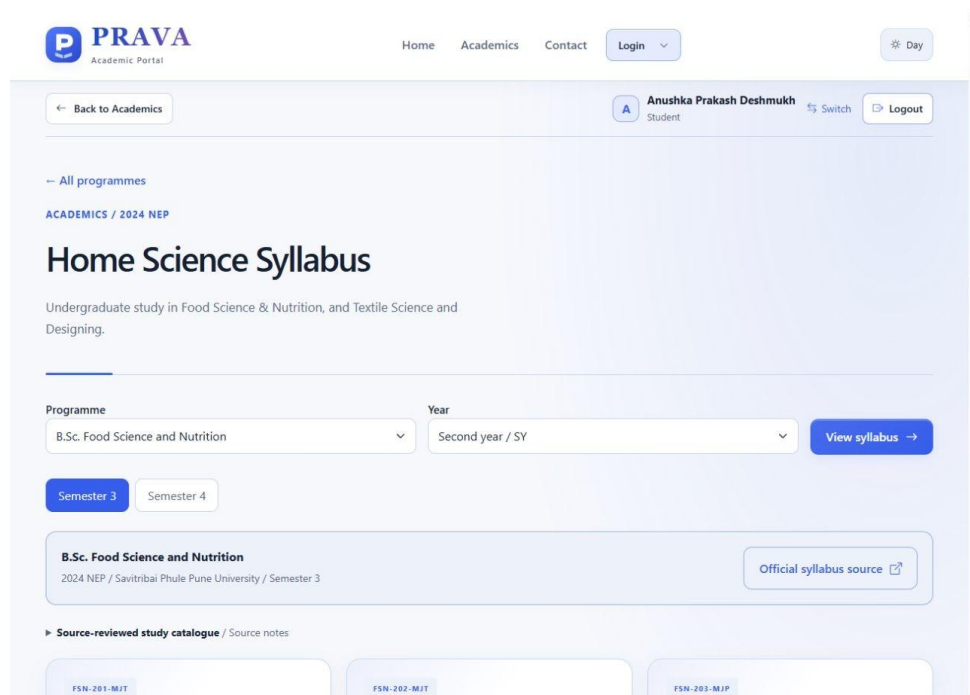


Figure 19 Home Science programme and semester catalogue

Anushka Prakash Deshmukh is configured for second-year Food Science and Nutrition. Home Science records also cover Textile Science and Designing, with FY, SY and TY students linked to their respective programmes.

Staff Dashboard and Attendance

The screenshot shows the 'STAFF WORKSPACE / OVERVIEW' page for Ms. Surekha K. Kale. The dashboard includes a sidebar with navigation links like 'Overview', 'My profile', 'My subjects', 'My students', 'Attendance', 'Attendance reports', 'Marks & grades', 'Marks reports', 'Study materials', 'Assignments', 'Announcements', 'College activities', and 'College home'. The main content area displays a welcome message and a summary of teaching responsibilities: 6 Assigned Subjects, 3 Students, 2 Active Assignments, and 0 Pending Reviews. Below this, there are sections for 'Assigned Subjects' and 'Active Assignments'. The 'Assigned Subjects' section lists three subjects: 'Problem Solving and Programming in C' (CA-101-T | BCA Semester 1), 'Laboratory: Problem Solving and Programming in C' (CA-102-P | BCA Semester 1), and 'Advanced C Programming' (CA-151-T | BCA Semester 2). The 'Active Assignments' section lists two assignments: 'Coursework: Problem Solving and Programming in C' (CA-101-T | Due 16 Oct 2026) and 'Coursework: Laboratory: Problem Solving and Programming in C' (CA-102-P | Due 16 Oct 2026).

Figure 20 Staff dashboard with teaching summary

The screenshot shows the 'ATTENDANCE MODULE' page for Ms. Surekha K. Kale. The page is titled 'Mark Attendance' and includes a 'Report' button. Below the title, there is a form to select a subject and a date. The subject is 'CA-101-T - Problem Solving and Programming in C' and the date is '28-09-2026'. A 'Load' button is next to the date field. Below the form, there is a table showing attendance for the selected subject and date. The table has columns for 'ENROLLMENT', 'STUDENT', 'STATUS', and 'REMARKS'. The table contains three rows of data, all with a status of 'Present' and a remark of 'Recorded in class'.

ENROLLMENT	STUDENT	STATUS	REMARKS
PRV-BCA-2026-001	Aditi Sandeep Shinde bca@prava.example	Present	Recorded in class
PRV-BCA-2026-008	Anagha Pravin Shinde anagha.shinde@prava.example	Present	Recorded in class
PRV-BCA-2026-009	Prajakta Sunil Lokhande prajakta.lokhande@prava.example	Present	Recorded in class

Figure 21 Attendance for an allocated teaching subject

Ms. Surekha K. Kale has six teaching allocations. The attendance screen loads the assigned BCA cohort and the selected class date. Academic entries and teaching allocations shown here are configured for the local project presentation.

Office Dashboard and Registration Review

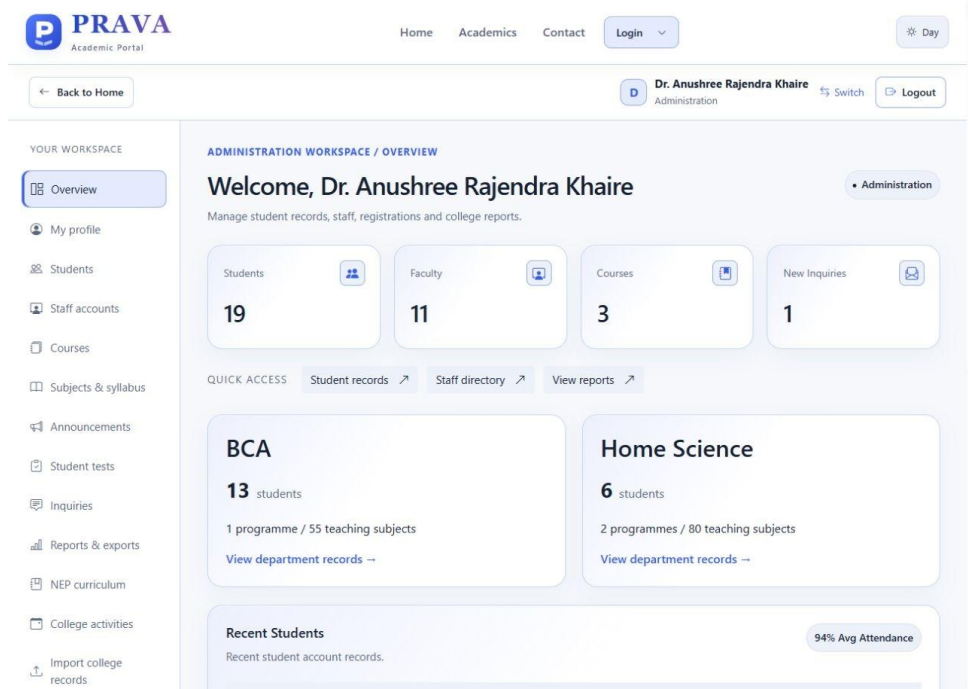


Figure 22 Office overview with BCA and Home Science departments

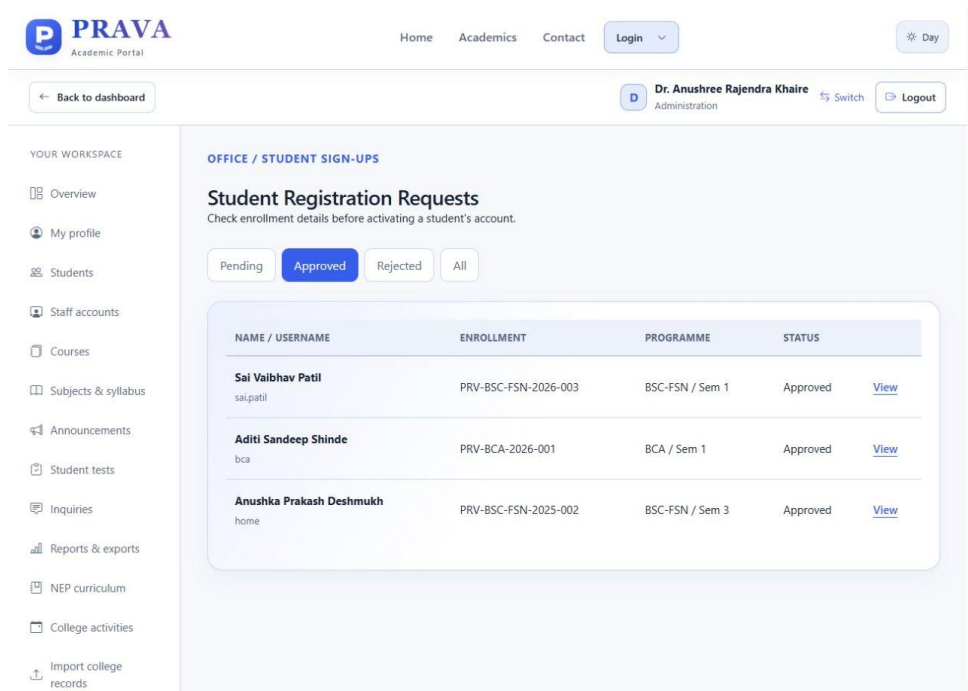


Figure 23 Office review queue for student sign-ups

The administration account displays Dr. Anushree Rajendra Khaire. The overview contains 19 student and 11 staff profiles, and the approved queue shows completed local registration records. Principal oversight also has a separate read-only account.

Roster Import and Report Outputs

PRAVA Academic Portal

Home Academics Contact Login

Dr. Anushree Rajendra Khaire Administration Switch Logout

Back to dashboard

YOUR WORKSPACE

- Overview
- My profile
- Students
- Staff accounts
- Courses
- Subjects & syllabus
- Announcements
- Student tests
- Inquiries
- Reports & exports
- NEP curriculum
- College activities
- Import college records

OFFICE / RECORDS

Import College Records

Validate the complete file, review the preview, then save the records together.

Start with a template

Use official enrollment numbers, employee IDs and email addresses. Programme codes: BCA, BSC-FSN, BSC-TEXTILE. Pattern: 2024 NEP.

Student CSV template Staff CSV template

New records stay inactive until the office sets an individual password and activates each account. Existing records are protected from duplicate imports.

Record type

Students

College roster (UTF-8 CSV, up to 200 rows)

Choose File No file chosen

Validate & preview

Figure 24 Separate Student and Staff CSV import templates

PRAVA Academic Portal

Home Academics Contact Login

Dr. Anushree Rajendra Khaire Administration Switch Logout

Back to dashboard

YOUR WORKSPACE

- Overview
- My profile
- Students
- Staff accounts
- Courses
- Subjects & syllabus
- Announcements
- Student tests
- Inquiries
- Reports & exports
- NEP curriculum
- College activities
- Import college records

REPORTS MODULE

Reports

Generate CSV reports for academic and master data.

Print

Student Report
Enrollment, contact, course, semester, and active status.
Download CSV

Faculty Report
Employee ID, department, contact, subject count, and active status.
Download CSV

Attendance Report
Daily attendance rows with student, subject, status, and faculty.
Download CSV

Marks Report
Subject marks, totals, grades, and pass/fail result.
Download CSV

Assignment Report
Assignments, due dates, maximum marks, and submission counts.
Download CSV

Notification Report
Announcements, target filters, creator, and active status.
Download CSV

Figure 25 Administrative report and CSV export options

The import workflow validates a complete file before commit. Reports and exports use the stored records and applicable access scope.

Student Marks and Coursework

PRAVA
Academic Portal

Home Academics Contact Login

Aditi Sandeep Shinde
Student

Back to dashboard

YOUR WORKSPACE

- Overview
- My profile
- Subjects & syllabus
- My attendance
- My marks**
- Study materials
- Assignments
- Announcements
- Student test
- My NEP syllabus
- My activities
- College activities
- College home

STUDENT WORKSPACE / RESULTS

My marks

Your academic results, one subject at a time.

Aditi Sandeep Shinde
PRV-BCA-2026-001 / BCA / Semester 1

7 results

Subject-wise results

Internal and external marks, totals, and grades recorded by your teaching staff.

SUBJECT	EXAM TYPE	INTERNAL	EXTERNAL	TOTAL	GRADE
Computer Organization & Architecture CA-103-T	Semester Exam	11	25	36 / 50	Pending
Discrete Mathematics and Statistics CA-105-T	Semester Exam	14	25	39 / 50	Pending
HTML and Web Page Designing SEC-101-CA	Semester Exam	11	25	36 / 50	Pending
Laboratory: Computer Organization & Architecture CA-104-P	Semester Exam	15	25	40 / 50	Pending

Figure 26 Student subject results with component marks

PRAVA
Academic Portal

Home Academics Contact Login

Aditi Sandeep Shinde
Student

Back to dashboard

YOUR WORKSPACE

- Overview
- My profile
- Subjects & syllabus
- My attendance
- My marks
- Study materials
- Assignments**
- Announcements
- Student test
- My NEP syllabus
- My activities
- College activities
- College home

STUDENT MODULE

My Assignments

View assignment details, submit work, and check grades.

TITLE	SUBJECT	DUE DATE	MAX MARKS	STATUS	MARKS	FEEDBACK	ACTION
Coursework: Problem Solving and Programming in C	CA-101-T	16 Oct 2026	20	Graded	16	Well organised work. Add clearer references in the next submission.	Download
Coursework: Laboratory: Problem Solving and Programming in C	CA-102-P	16 Oct 2026	20	Graded	16	Well organised work. Add clearer references in the next submission.	Download

Figure 27 Submitted coursework with marks and faculty feedback

The student results page shows stored subject components and totals. Official NEP grade calculations remain pending. Coursework lists due dates, submission status, awarded marks and feedback for the signed-in student.

Staff Review and Grading

PRAVA Academic Portal

Home Academics Contact Login

Ms. Surekha K. Kale Staff Logout

← Back to assignments

YOUR WORKSPACE

- Overview
- My profile
- My subjects
- My students
- Attendance
- Attendance reports
- Marks & grades
- Marks reports
- Study materials
- Assignments**
- Announcements
- College activities
- College home

ASSIGNMENT MODULE

Coursework: Problem Solving and Programming in C

CA-101-T / Due 16 Oct 2026, 05:00 PM

STUDENT	ENROLLMENT	STATUS	SUBMITTED	MARKS	FEEDBACK	ACTION
Aditi Sandeep Shinde	PRV-BCA-2026-001	Graded	03 Oct 2026, 10:00 AM	16	Well organised work. Add clearer references in the next submission.	Download Edit
Anagha Pravin Shinde	PRV-BCA-2026-008	Graded	03 Oct 2026, 10:00 AM	16	Well organised work. Add clearer references in the next submission.	Download Edit
Prajakta Sunil Lokhande	PRV-BCA-2026-009	Graded	03 Oct 2026, 10:00 AM	17	Well organised work. Add clearer references in the next submission.	Download Edit

Figure 28 Faculty review of submissions for an allocated subject

PRAVA Academic Portal

Home Academics Contact Login

Ms. Surekha K. Kale Staff Logout

← Back to dashboard

YOUR WORKSPACE

- Overview
- My profile
- My subjects
- My students
- Attendance
- Attendance reports
- Marks & grades
- Marks reports
- Study materials
- Assignments**
- Announcements
- College activities
- College home

ASSIGNMENT MODULE

Grade Submission

Aditi Sandeep Shinde / Coursework: Problem Solving and Programming in C

Marks Obtained

16

Maximum marks: 20

Feedback

Well organised work. Add clearer references in the next submission.

Cancel Save Grade

Figure 29 Assignment score and feedback entry

The assigned faculty member reviews the submitted file, awards a score within the assignment maximum, and records feedback. The displayed 16 out of 20 and review text are existing local presentation records.

11 Testing and Verification

All 63 automated tests passed on 5 October 2026 in 26.480 seconds using isolated fixtures. The run used repository baseline 4abb237 with local typography improvements. The controlled error-route test intentionally logs an exception while checking the friendly error response.

Test module	Tests	Result
college_upgrade	15	Passed
onboarding	11	Passed
phase14	12	Passed
public_homepage	20	Passed
student_test	5	Passed
Total	63	Passed

Representative case	Expected and verified result
Login and switching	Role/programme checks; matching dashboard; failed switch preserves prior identity.
Registration and imports	Pending review; duplicate conflicts rejected; transaction rollback prevents partial accounts.
Permissions and security	Principal write restrictions; Staff subject ownership; CSRF and inactive account checks.
Academics and communication	NEP score bounds; curriculum publication status; notice audiences; activity ownership.
Public pages and student tests	Separate pages, valid contact storage, protected student tests and correct score handling.

Browser checks on 5 October 2026 confirmed all four main login flows and refreshed the report screenshots. Day and Night layouts were inspected at desktop and mobile widths. Key labels, form fields and table text use larger type; wide tables remain horizontally scrollable within their panels.

Input Validation Examples

The following examples explain implemented safeguards. They were checked through the isolated automated test suite on 5 October 2026; no invalid records were added to the working college database.

Login and programme checks

An incorrect password returns “Invalid username/email or password.” A BCA account cannot sign in through the Home Science portal. Inactive accounts are blocked, and an unsuccessful account switch preserves the previous valid identity.

Duplicate registration and import

Submitting the same student sign-up twice leaves one request. An existing enrollment / employee number is rejected by the CSV import preview. Approval and import commit repeat uniqueness checks, so a conflicting record prevents partial account creation.

Marks and assignment bounds

Validation example	Result
Internal maximum 15; enter 16	Rejected: component exceeds its limit.
External maximum 35; enter 36	Rejected: component exceeds its limit.
Assignment maximum 100; enter -1 or 101	Rejected: negative or above-maximum score.
Student from a different academic cohort	Rejected: no eligible subject membership.

Verification evidence

test_phase14 covers login and assignment bounds; test_onboarding covers repeated sign-up and approval conflicts; test_college_upgrade covers NEP component limits, roster duplicates and curriculum ownership. All 63 tests passed in 26.480 seconds.

12 Security and Data Maintenance

Access and consistency

Passwords are stored as hashes. Protected routes check authentication, active status, roles and academic ownership. State-changing requests require CSRF tokens, including Logout. Imports and registration approval revalidate duplicates before commit. Unique keys and foreign keys complement service-level checks; selected actions are recorded in activity_logs.

Persistent storage and recovery

Location	Stored information
instance/prava.sqlite3	Users, profiles, academic records, inquiries, review and activity data.
instance/.local-secret	Persistent local session secret.
instance/backups/	Backups created by the launcher before detected schema upgrades.
app/static/uploads/	Profile images, materials, assignments and submitted work.
Google response Sheets	Student and Staff intake responses; separate from the PRAVA database.

A complete backup preserves the database and required uploaded files. Stop the local server for a file-copy backup or use a consistent database backup procedure. Reopening the same installation retains its records. A newly extracted ZIP creates a separate installation, so populated folders should be preserved during updates.

Public deployment

The launcher binds to 127.0.0.1. Public hosting requires production settings, HTTPS, persistent database and file storage, private secrets, restricted administration and tested recovery. Local presentation credentials must be disabled for real operation. Repository publication delivers source code; it does not publish the local database or deploy a public service.

13 Advantages Limitations and Future Work

Advantages

- Public information and academic work are available through one consistent portal.
- Programme-aware student access and explicit Staff allocations keep records in the correct academic context.
- Office review, duplicate checks and transactional imports support controlled account creation.
- Source links, review labels, persistent local storage and CSV exports support demonstration and record review.

Current limitations

The catalogue has 131 entries: 123 source-reviewed public entries and eight drafts. Shared Home Science content appears under both specialisations. The BCA-linked source is titled B.Sc. Computer Applications; SY is marked draft. Twelve local Home Science TY practice topics support presentation without being labelled verified university syllabus. Detailed TY sources and some assessment details await confirmation.

Official NEP grade, SGPA/CGPA and individual elective enrolment are not implemented. The project does not automate email after Office approval. The Google Forms require review and manual conversion to import templates. The small local presentation dataset does not establish institutional capacity or full-semester operation.

Future enhancements

Priorities are approved batch-specific syllabus updates, elective registration, institution-validated grade calculations, controlled account recovery and registration notifications. Hosted operation should add verified backup restore, accessibility review and performance testing with representative data.

14 Project Cost and Conclusion

Illustrative cost estimate

The following is an academic planning example in INR, not recorded expenditure or logged effort. It assumes an existing laptop and connectivity. Team-hours are the combined effort of both members and are not multiplied by two again.

Planning item	Calculation	Value INR
Requirements and design	40 hours $\times$ 100	4,000
Implementation	70 hours $\times$ 100	7,000
Verification	26 hours $\times$ 100	2,600
Documentation	24 hours $\times$ 100	2,400
Total effort value	160 hours $\times$ 100	16,000
Printing budget	Up to 45 pages $\times$ 3 $\times$ 2 copies	270
Binding budget	150 $\times$ 2 copies	300
Local hosting fee	No paid hosting for local presentation	0
Illustrative combined value	16,000 + 270 + 300	16,570

Conclusion

PRAVA demonstrates a working college portal with clear role entry, programme-aware student access, Office-reviewed onboarding and academic tools. The relational design separates identity, teaching allocation, curriculum references and academic records. The current interface, 22-table schema, portable launcher and 63 passing tests support the implementation described in this report.

The project provides practical experience in web development, relational modelling, validation and software verification. Further work should complete approved academic rules and source review, then evaluate real operational data and hosting requirements.

15 Bibliography

Project implementation and technical references. The software versions are the repository requirements; curriculum references are preserved from the application source review.

1. PRAVA source repository

kvaishu0703/Prava_Edu. Baseline 4abb237 with local typography improvements; dataset, interface and 63 automated tests verified on 5 October 2026. README, models, services, templates and tests.
https://github.com/kvaishu0703/Prava_Edu

2. Python documentation

Python Software Foundation. Python 3.12 documentation.
<https://docs.python.org/3.12/>

3. Flask documentation

Pallets Projects. Application factories, configuration, blueprints and testing. Project dependency Flask 3.0.3.
<https://flask.palletsprojects.com/en/stable/>

4. SQLAlchemy ORM documentation

SQLAlchemy. Object relational mapping and model relationships.
<https://docs.sqlalchemy.org/en/20/orm/>

5. College information

Women's College of Home Science and BCA, Loni. Institutional information and academic references recorded by the project.
<https://www.pravarahomesciencebca.in/>

6. Google Forms setup

Google Apps Script Form service. Separate student/staff forms, validation, publishing and spreadsheet destinations; created 30 September 2026 and updated to English on 4 October 2026.
<https://developers.google.com/apps-script/reference/forms/form>

Curriculum Source References

7. SPPU FY B.Sc. Computer Applications syllabus linked by the college BCA department.

http://www.pravarahomesciencebca.in/documents/STUDENT%20CORNER/Syllabus/BCA/B.%20Sc.%20%28Computer%20Applications%29_14062024.pdf

8. SPPU SY B.Sc. Computer Applications draft hosted by Pratibha College.

https://pcccs.org.in/wp-content/uploads/2026/03/SY-B.Sc_.CA_.pdf

9. SPPU FY B.Sc. Home Science syllabus linked by the college.

http://www.pravarahomesciencebca.in/documents/STUDENT%20CORNER/Syllabus/Home%20Science/syllabus/B.Sc.%20%28Home%20Science%29_07062024.pdf

10. SPPU SY B.Sc. Home Science syllabus on the university circulars site.

[http://collegecirculars.unipune.ac.in/sites/documents/Syllabus2025/S.Y.B.Sc.%20\(Home%20Science\)_25082025.pdf](http://collegecirculars.unipune.ac.in/sites/documents/Syllabus2025/S.Y.B.Sc.%20(Home%20Science)_25082025.pdf)

The catalogue records its source-check date as 20 September 2026. Source title, pattern, effective year and review notes must be interpreted together. A college department label may differ from a university document title. Formal academic use requires confirmation that each document applies to the student batch.

The report preserves known source qualifications and current implementation status. It does not convert draft or incomplete syllabus content into a confirmed university scheme.

Appendix Local Installation and Demonstration

- Open https://github.com/kvaishu0703/Prava_Edu and choose Code → Download ZIP, then Extract All.
- Preserve any existing populated project folder. A new folder has its own local database.
- Install supported 64-bit Python; the README recommends 3.12 with Add Python to PATH.
- Open the inner folder containing START-PRAVA.cmd and double-click it. Internet is needed for first dependency setup.
- Keep the server window open. The browser opens <http://127.0.0.1:5000/>. Stop the server with Ctrl+C.

PowerShell: right-click the project folder → Open in Terminal. Run `.\START-PRAVA.cmd`. If port 5000 is busy, run `.\START-PRAVA.cmd -Port 5001` and open <http://127.0.0.1:5001/>. The `.\` prefix selects the file in the current folder.

Role	Login ID	Password
BCA student	bca	bca123
Home Science student	home	home123
Staff	staff	staff123
Office / Administration	office	office123

These credentials are for local presentation only. Verified college accounts use individual credentials. The local accounts are blocked when `PRAVA_DEMO_MODE` is disabled or production configuration is active.

Presentation order: Home and theme → Login dropdown → BCA and Home Science dashboards → Staff academic tools → Office review, import and reports. Use Logout or Switch account between roles. README.md contains the setup and maintenance instructions.